

DMAIC PROJECT CHARTER

Flight Delay Root Cause Analysis | US Domestic Flights 2023

Problem Statement

US domestic flights in 2023 achieved only ~78% on-time arrival rate — meaning 1 in 5 flights arrived delayed. Root causes are not systematically identified, measured, or monitored across the industry.

Goal Statement

Identify the top 2-3 root causes of US domestic flight delays using 2023 BTS data, quantify their sigma-level impact, and design a KPI monitoring dashboard to enable data-driven operational decisions.

Scope

IN SCOPE:

OUT OF SCOPE:

- All US domestic flights, full year 2023
- All reporting carriers
- All delay cause categories (carrier, weather, NAS, security, late aircraft)

- International flights
- Airport ground crew operations

CTQ Metrics

Primary Metric:

On-Time Arrival Rate (%) — flight on-time if arrives within 15 minutes of scheduled arrival

Secondary Metric:

Average Arrival Delay (minutes)
DPMO & Sigma Level

DMAIC Timeline

DEFINE

MEASURE

ANALYZE

IMPROVE

CONTROL

Jun 18-24 → Jun 25-Jul 8 → Jul 9-22 → Jul 23-Aug 5 → Aug 6-12

Tools & Methodology

Framework: DMAIC (Lean Six Sigma Green Belt)

Database: SQL — SQLite / DB Browser

Visualize: Tableau Public

Dashboard: Microsoft Power BI

Statistics: Excel (t-test, Regression, Cp/Cpk)

Data Source: BTS On-Time Performance 2023

Owner: Sabab Abrar Hamim

Cert: Lean Six Sigma Green Belt (June 2026)

University: UT Arlington — Industrial, Manufacturing, and Systems Eng.

Minors: Information Systems | Sustainable Eng.

Timeline: June – August 2026

